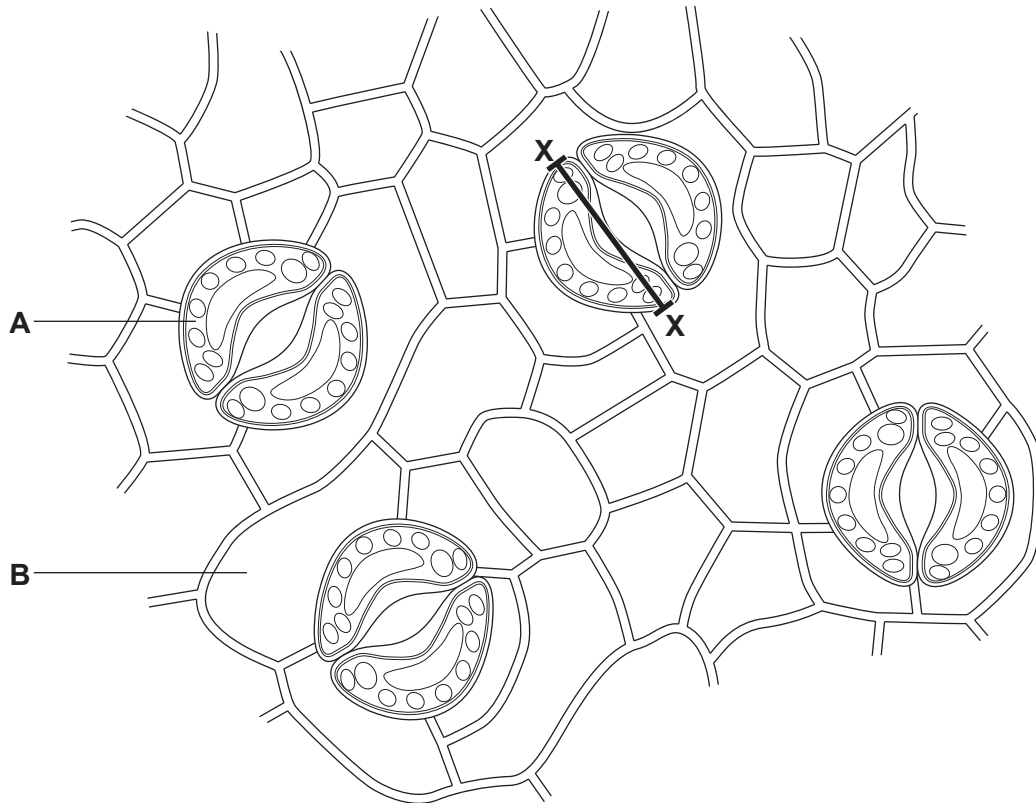


5. **Image 5.1** shows an illustration of part of the lower surface of a leaf from a cress plant.

Image 5.1



- (a) (i) State the names of the type of cells labelled **A** and **B** in **Image 5.1**. [2]

A

B

- (ii) Measure the length of line **X–X** on **Image 5.1**. [1]

Length = mm

- (iii) The actual length of the cell measured by line **X–X** on **Image 5.1** is 20 μm . Calculate the magnification of the drawing. [2]

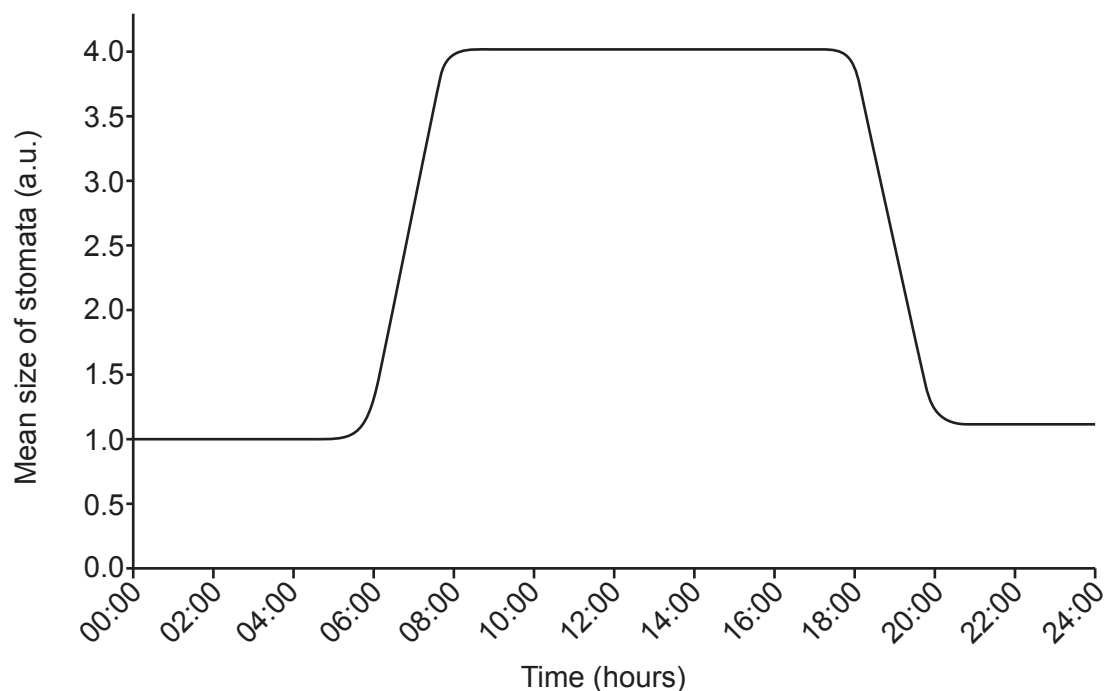
(1 mm = 1000 μm)

Magnification = $\times$



An investigation was carried out into the changes in the mean size of the stomata of a cress plant. The plant was left on a windowsill and the changes were recorded over a 24-hour period. The results are shown in **Graph 5.2**.

Graph 5.2



- (b) (i) State **one** conclusion that can be drawn from **Graph 5.2** about when stomata open and close. [1]

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.....

- (ii) Explain **one** advantage to plants of being able to change the size of their stomata. [1]

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